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A Matched Filter for Cosmic Ray Detection from Electromagnetic Wave Reflection

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A new technique for detecting high-energy cosmic rays by using the front scattering of electromagnetic waves has been introduced recently. The reflected signals are captured by antennas and demodulated. The present work makes use of optimal filtering for detecting cosmic rays and removing the huge background noise. The filter detecting system achieves 98% of the detection efficiency for a false alarm probability as low as 1.0%.

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1. Introduction

High energy cosmic rays are strongly accelerated nucleons that reach a kinetic energy higher then $10^{20} eV$ [1]. This natural acceleration process became one of the main issues in physical research, given rise to the construction of modern instruments to detect and study cosmic rays [2].

Recently [3], for detecting extremely-high energy cosmic rays, a new detection technique has been introduced, which uses the front scattering of electromagnetic waves [4]. This method has extensively been used for meteor detection [5] and it can be adapted to cosmic ray detection requirements. As meteors, high energy cosmic rays are able to ionize atoms from the atmosphere. This ionization process reflects on electromagnetic waves that are captured by antennas and demodulated. Due to the weak intensity of the reflected wave, signal detection has to be performed in low signal-to-noise ratio conditions. Hence, stochastic signal processing techniques look attractive, as they are able to extract signals from noise [6].

In this work, the optimal filtering technique [7] is applied to online detection of cosmic rays from electromagnetic wave reflection. The model assumes that the received signal is masked by additive noise and both signal and noise statistics are used to design a linear filter that maximizes the signal-to-noise ratio. This filter is known as the matched-filter [8], as detection is performed by correlation of the incomming signal with replicas of the target signal components in the receiver end. The detection performance of the online proposed detection system was evaluated using experimental digitized data.

The paper is organized as it follows. An overview of the cosmic ray detection system and data acquisition setup presented in the next two sections. Then the detection theory is summarized in Section IV. The signal and noise characterization and the detection efficiency of the system are given in Section VI. Section VII derives some conclusions.

2. Overview

Figure 1 shows the cosmic ray detection principle. Primary cosmic rays generate a shower of particles in the atmosphere and charged particles from this shower ionize the atmospheric gas. The free electrons from this ionized gas reflect on electromagnetic waves coming from a parasitic transmitter antenna. A receptor antenna, tuned to the transmitter carrier, captures waves reflected by the atmosphere when a cosmic shower arrives. Thus, the cosmic shower detection is obtained indirectly. In order to get only reflected waves and to avoid signals coming directly from the transmitter, the receptor antenna is positioned far away from the transmitter. This technique allows to detect not only cosmic rays but any kind of physical phenomenon that is able to ionize the atmospheric gas, as meteors, lightning and airplanes. Eventually, scintilators can be used to detect cosmic rays in a small area. This is useful for the calibration and test of the detector setup [9].

The forward scattering of radio waves by ionized gas is a well known physical phenomenon described by classical electromagnetism theory [4]. One should use VHF (Very High Frequency) transmitter stations in the range from 30 to 300 MHz positioned far away from the receptor station. With this approach the direct signal from the transmitter does not reach the receptor, since the antenna is bellow the horizon line and also there is no reflection in the ionosphere for this range of

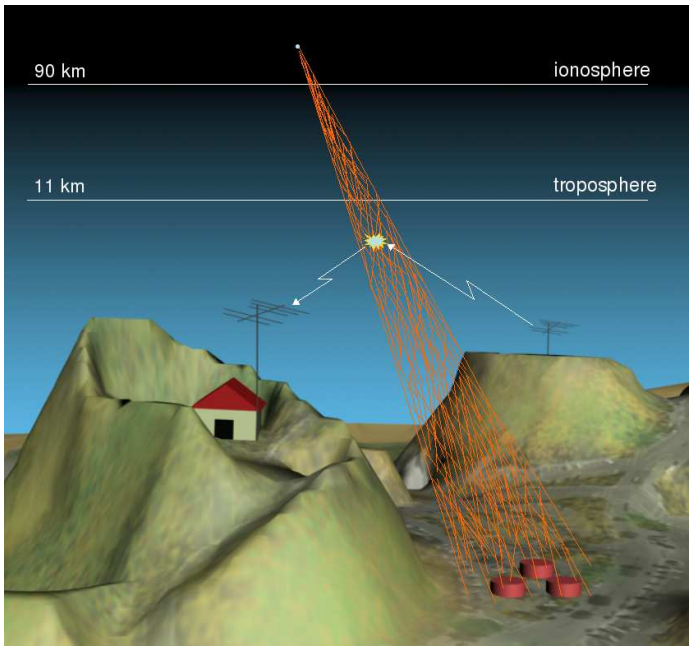

Figure 1: Cosmic ray detection for scattering of electromagnectic waves.

frequency. In order to have reflection of the VHF waves is necessary that the electronic density of the ionized gas be higher than a threshold depending of the frequency of these waves.

3. Data Acquisition System

Data acquisition is still in an initial phase. A prototype assembled with off-the shelf parts is being use. Figure 2 shows the system used for the first experimental data. The radio receivers are set to the TV commercial channels 55.25 and 67.25 MHz, the distance from the transmitters being enough to get only reflected waves. The signal coming from the scintilators are acquired using standard NIM crate. A GPS (Global Positioning System) is used in order to synchronize the information when there is more than one receptor antenna.

All analog signals are digitized in a 24-bit word by an 8-channel commercial sound board. The sampling frequency can be performed in the range 8 to 96 kHz. The sound board is connected to a PC (Personal Computer), which runs the software that acquires and stores data in huge volumes (mostly background noise).

A typical signal generated by a cosmic shower is showed in Figure 3. Due to the natural randomicity of the cosmic shower signals, they can be considered as a stochastic process. To increase the detection efficiency for events of interest, statistic methods of the signal detection were used. These methods, known by matched-filter, are used when the signal-to-noise ratio is low, but the stochastic characteristics of the signal and of the additive noise are known. Besides, since different types of phenomenon can reflect electromagnetic wave in the atmosphere, simpler detection methods as patamar decision are not enough and a classification technique is necessary. However, matched-filters are also a linear classification method in the sense that only signals of interest are detected.

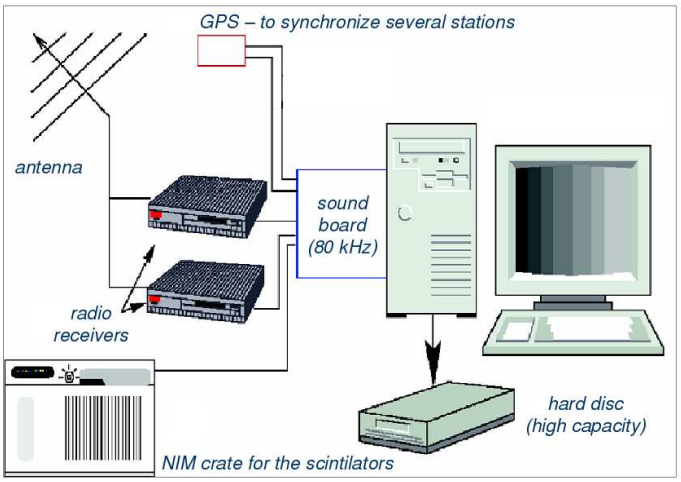

Figure 2: Data Acquisition System.

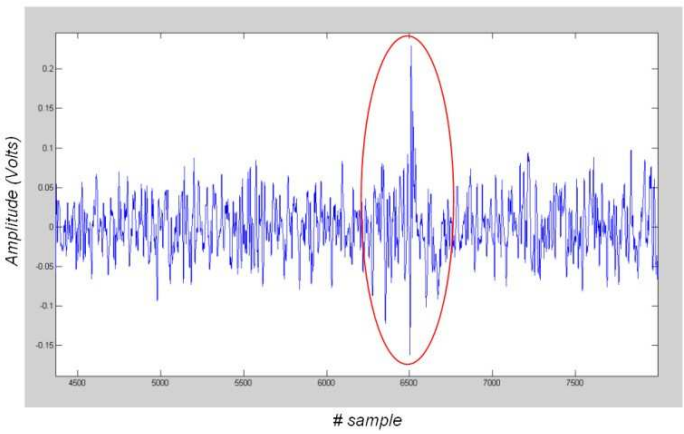

Figure 3: Typical signal potentially generated by a cosmic shower.

The proposal of the present work is to implement a free-running algorithm to detect cosmic rays using the matched-filter technique. This algorithm should be able to run online in the next versions of the setup, in order to store only event of interest. As a first approach and to keep the algorithm as fast as possible, a matched-filter for deterministic signal was implemented using the mean signal of the stochastic process. The theory of the detection of the deterministic signals in systems with low signal-to-noise ratio is presented in the next section.

4. Detection Theory

Given a discrete signal $s[k]$ with N samples to be transmitted in a medium which inserts an additive noise, one would like to determine, as more efficiently as possible, if the signal is really present. In other words, a processing should be done in order to decide between two possible hypotheses: only the noise is present (called H_0) or both the desired signal and the additive noise are present (called H_1). Obviously, the decision between H_0 and H_1 should be based on the hypothesis

with higher probability for an observed segment of N samples. This decision rule is summarized by the Equation 4.1

$$\frac{P(H_1|\mathbf{l})}{P(H_0|\mathbf{l})} \underset{H_0}{\overset{H_1}{\gtrless}} 1 \tag{4.1}$$

where $P(H_1|\mathbf{l})$ and $P(H_0|\mathbf{l})$ are the conditional probabilities of the H_1 and H_0 respectively, for a given outcome of N samples. The Equation 4.1 is interpreted as following: one decides by H_1 if the probability ratio is higher than 1, otherwise H_0 is chosen. Using the Bayes rule for conditional probabilities

$$P(H_i|\mathbf{l}) = \frac{f_{\mathbf{L}|H_i}(\mathbf{l}|H_i)P(H_i)}{f_{\mathbf{L}}(\mathbf{l})} \tag{4.2}$$

we have

$$\frac{P(H_1)\, f_{\mathbf{L}|H_1}(\mathbf{l}|H_1)}{P(H_0)\, f_{\mathbf{L}|H_0}(\mathbf{l}|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} 1 \tag{4.3}$$

or

$$\frac{f_{\mathbf{L}|H_1}(\mathbf{l}|H_1)}{f_{\mathbf{L}|H_0}(\mathbf{l}|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{P(H_0)}{P(H_1)} \tag{4.4}$$

The left ratio is called the Likelihood ratio. The right ratio of the Equation 4.4 can vary, depending on the costs in deciding between one of the two hypothesis. A more general decision rule is given by Equation 4.5

$$\frac{f_{\mathbf{L}|H_1}(\mathbf{l}|H_1)}{f_{\mathbf{L}|H_0}(\mathbf{l}|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma \tag{4.5}$$

If the additive noise is white (decorrelated samples) and gaussian, the samples are statistically independent and the probability density functions are the productorials of the probability of each sample

$$\frac{\prod_{i=1}^{N} f_{L_i|H_1}(l_i|H_1)}{\prod_{i=1}^{N} f_{L_i|H_0}(l_i|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma \tag{4.6}$$

for a zero mean noise, we have

$$\frac{\prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma}} exp\{-\frac{(l_i - s[i])^2}{2\sigma^2}\}}{\prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma}} exp\{-\frac{l_i^2}{2\sigma^2}\}} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma \tag{4.7}$$

applying the logarithm in Equation 4.7 and rearranging the terms, we have

$$\mathbf{l}^T\mathbf{s} \underset{H_0}{\overset{H_1}{\gtrless}} \sigma^2 ln\gamma + \frac{1}{2}(\mathbf{s}^T\mathbf{s}) \tag{4.8}$$

The parcels on the right of the Equation 4.8 are constants values and they can be interpreted as a new patamar value γ'

$$\mathbf{l}^T\mathbf{s} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma' \tag{4.9}$$

This Equation show us that if the noise is gaussian and white, the decision between H_0 e H_1 is given by the internal product between the received signal and a copy of the signal of interest. The patamar of decision depends on the relationship between the probability of detection and the allowed false alarm. This kind of processing is called the matched-filter.

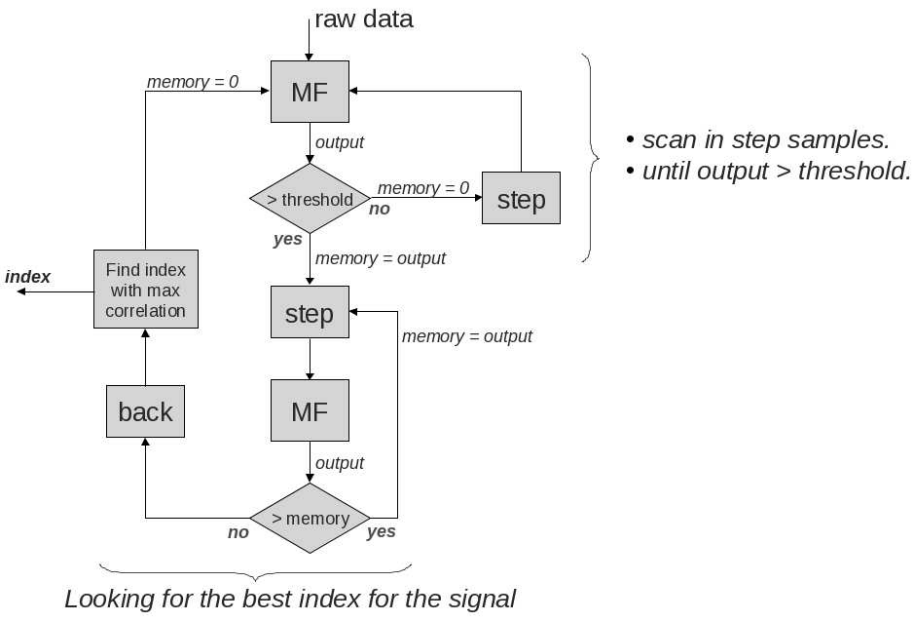

Figure 4: Free-running algorithm block diagram.

The matched-filter is a linear transformation which tries to increase the signal-to-noise ratio, using previous acknowledgment of the characteristics of the desired signal. It can be shown [7] that the matched-filter not only increase the signal-to-noise ratio but maximize its value, optimizing the detection efficiency. For this reason it is called an optimal filtering.

5. Free-running Algorithm

In order to use the matched-filter technique in the raw-data coming from the data acquisition software, a free-running algorithm was proposed (Figure 4). This algorithm is composed by two decision stages with matched-filters. The first stage applies a deterministic matched-filter at each *step* samples eliminating most of the noise and selecting cosmic candidates to be checked by the second stage. The optimal values for the *step* and the *threshold* for this stage were chosen using an artificial raw-data where the cosmic position was known, as explained in the Results section. In the second stage a more accurate decision, with a matched-filter applied at each sample, is performed. In this stage, pre-processing computations, like whitening and projection in a decorrelated base, can be performed to increase the detection efficiency.

6. Results

6.1 Noise Analyses

Figure 5 shows a zoom of the raw-data digitized where only the noise is present. A gaussian fit with the histogram computed with this data is showed in Figure 6. For a first approximation, the noise can be considered gaussian and the matched-filter technique presented can be used.

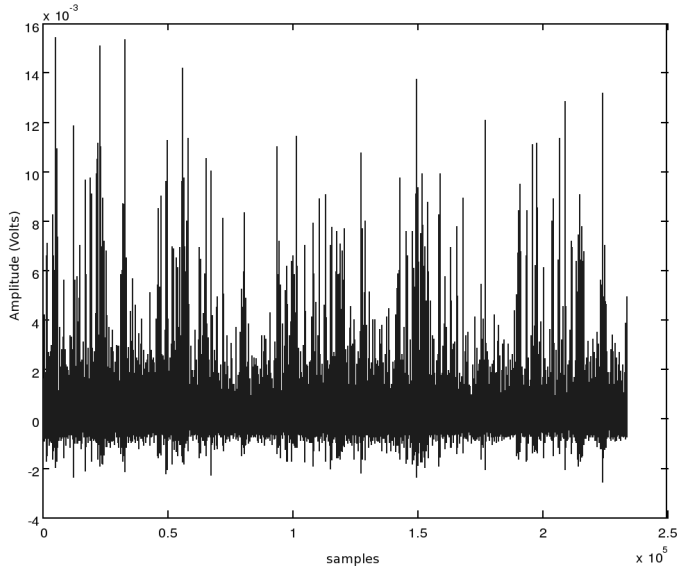

Figure 5: Raw-data segment only with noise.

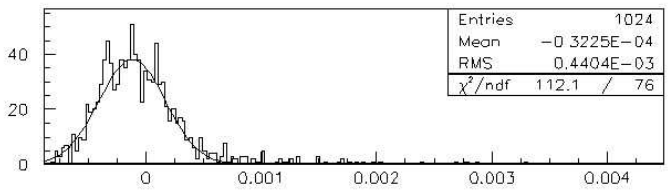

Figure 6: Gaussian fit applied to noise distribution.

The covariance matrix of the noise, showed in gray scale on Figure 7, is not completely diagonal. Due the low intensity of the non diagonal components, the noise can be considered white, for a first approximation, but a whitening preprocessing can be done in order to increase the detection efficiency.

6.2 Free-running parameters estimation

Potential preselected cosmic signals and noise were concatenated in a fake raw-data vector where the actual index of the cosmic signals are known. In this way, the detection probability versus the *threshold* and *steps* parameters can be plotted (Figure 8). One should choose combination of these two parameters where detection probability is higher them 100%, where the percentage of detection above 100% means the amount of noise erroneously detected as signal (false alarm) by the first level filter. These noises will be filtered by the second stage where a more accurate detection is performed whith a matched-filter applied at each sample around the *index* of the raw-data given by the first stage.

6.3 Detection efficiency

In order to determine the efficiency of detection, 480 preselected cosmic signals and 3600 noise segments were divided in two sets of the same length - the development and the test sets.

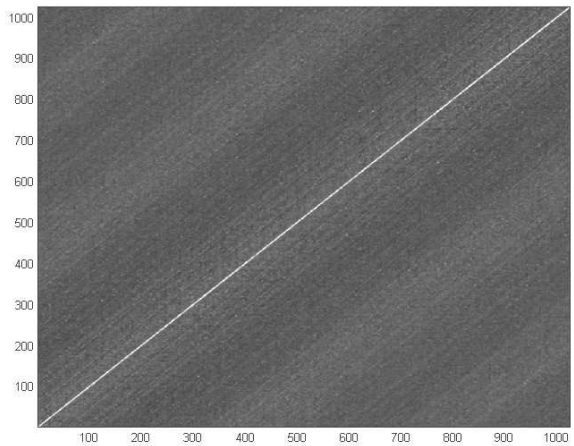

Figure 7: Covariance matrix of the noise in gray scale

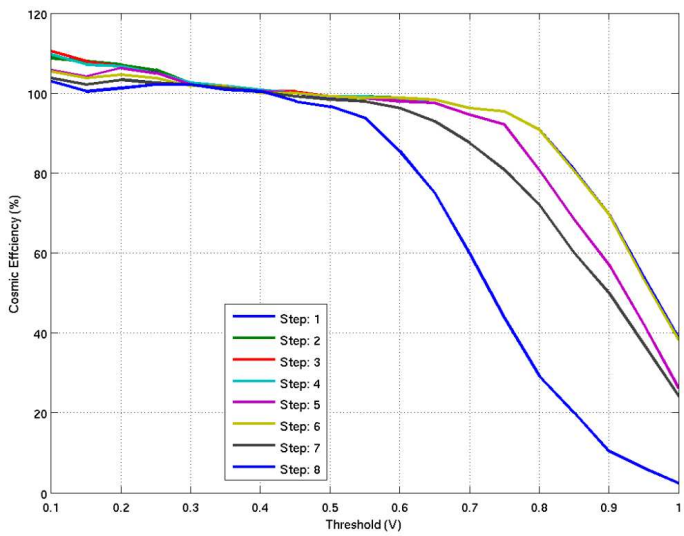

Figure 8: Detection efficiency versus *step* and *threshold* parameters

Figure 9 shows typical and mean pulse shapes for the noise and the cosmic signals using the development set.

Two fake raw-data were assembled using the development and test sets respectively. Figure 10 shows the detection efficiency versus the false alarm for these two sets. The signal used by the deterministic matched-filter was the mean signal of the development set. The blue line was done applying the matched-filter to each signal and noise separately. The red one is using the free-running algorithm in the fake raw-data. The value for the *threshold* and *step* parameters used were 0.5 and 6 respectively. Obviously the free running algorithm has an efficiency lower then the ideal case where the signals were previously separated, but for an accepted false alarm higher then 2.3%, the detection efficiency is 100% in both cases.

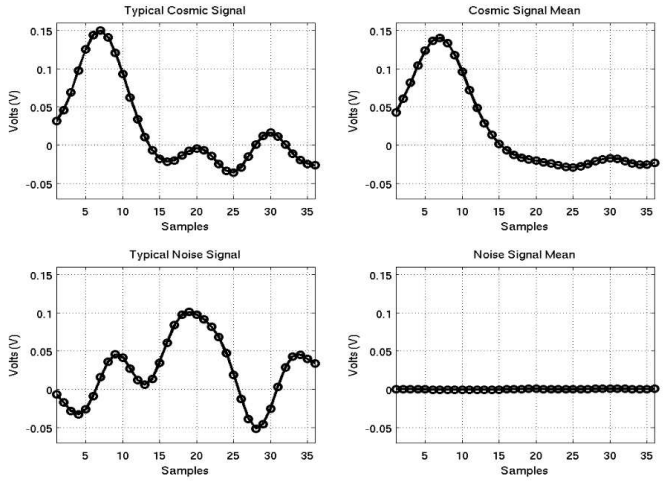

Figure 9: Typical and mean pulse for the cosmic and noise signals - development set

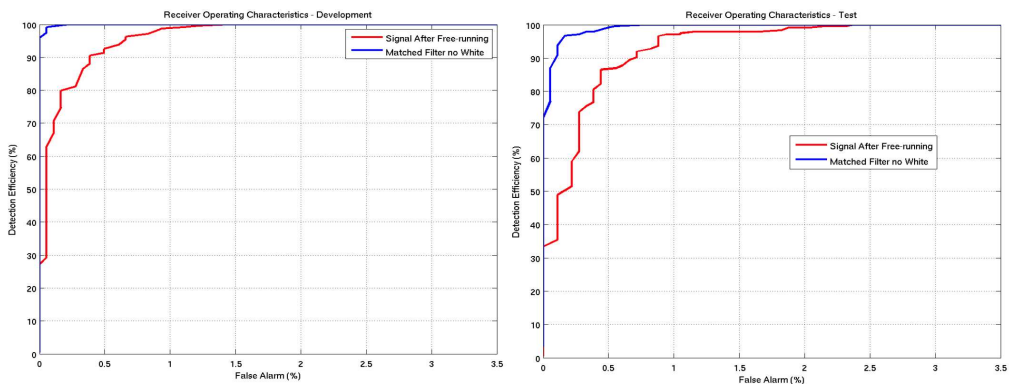

Figure 10: ROCs. Blue line - separated signals, red - free running algorithm

7. Conclusion and next steps

To improve the cosmic shower detection by using the front scattering of electromagnetic waves techniques, a deterministic matched filter using the mean signal of the cosmic signal stochastic process has been implemented. This filter achieves an efficiency of the 98% with a false alarm of the 0.5% for a preselected signal and noise test set. In order to use this filter in the raw-data generated by the detector setup, a free running algorithm was proposed achieving 1% of false alarm under the same detection probability (98%). Its two filtering stages characteristic allows an online utilization of this algorithm reducing drastically the amount of stored data.

Since the first stage of the free running algorithm eliminates most of the background noise and the second stage can run in a lower rate, a more accurate detection process can be performed by the last stage increasing the detection efficiency. Among them a whitening preprocessing using the covariance matrix of the noise and a projection in a decorrelated base, using the stochastic characteristic of the cosmic signal will be developed.

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